

Chavva Manoj

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Summary — Full Stack engineer with 2+ year shipping production systems at scale – built payroll infrastructure processing **1,900+ salaries in under 20 seconds**, a cost-optimized multi-modal AI platform achieving **82% retrieval precision at 65% lower inference cost**, and 3 end-to-end products across **Spring Boot, React/TypeScript**, and cloud-native stacks.

Skills / Tech Stack

Backend	Python, FastAPI, Java, Spring Boot, Microservices, REST APIs, JPA/Hibernate, JWT/OAuth2	Messaging	RabbitMQ, Kafka
Frontend	React.js, TypeScript, Next.js, JavaScript, HTML, CSS, Redux/Zustand	DevOps	Git, CI/CD, Docker, Vercel, Railway
Databases	PostgreSQL, MongoDB, Query Optimization	Cloud	AWS, Azure
		AI / LLM	RAG, LangChain, OpenAI APIs, FAISS, Prompt Engineering, Whisper
		Core	DSA, System Design, Distributed Systems

Experience

CDAC (Centre for Development of Advanced Computing)

Jul 2024 – Present

Project Associate

- **Processed payroll for 1,900+ employees in under 20 seconds** by engineering an automated salary disbursement pipeline using **schedulers** and **message queues**, eliminating manual processing bottlenecks and ensuring on-time, error-free payroll execution at scale.
- **Shipped 6 production microservices** handling payroll, billing, and sanctions modules by architecting a modular **Spring Boot + REST API** backend with **JWT-based auth**, reducing inter-service coupling and enabling independent scaling.
- **Reduced financial query latency by 3x** for billing, payroll, and sanctions workflows by redesigning **PostgreSQL** schema with **composite indexing**, **query optimization**, and **JPA/Hibernate** tuning across large-scale enterprise data.
- **Led delivery of Phase-1 enterprise platform** on schedule across a **4-developer Agile team** by driving end-to-end requirement gathering, sprint planning, and technical coordination — reducing stakeholder review cycles by consolidating 2 departments into a single deployment workflow.
- **Achieved zero-downtime migration of 3,000-employee financial records spanning 20+ years** by architecting a phased **data migration strategy** with rollback checkpoints and validation gates at each stage.
- **Eliminated single points of failure across a 3,000-employee production platform** by designing **multi-VM deployment topology** with failover routing, VM-level environment structuring, and **deployment lifecycle management** across dev, staging, and production.
- **Reduced deployment cycle to weekly releases** across 3 environments by building end-to-end **CI/CD pipelines** (Git → build → deploy) with Git-based versioning, release governance, and automated environment promotion.

Projects

Cost-Optimized Multi-Modal RAG Platform (FastAPI + React/TypeScript + Whisper)

Feb 2026 -Present

- **Reduced inference cost by 65% and achieved 82% retrieval precision** by building a production **RAG platform** with a **3-tier model router** (gpt-4o-mini / gpt-4o / gpt-4.1) that classifies query complexity and routes 60%+ of queries to the lowest-cost model.
- **Built the full-stack interface end-to-end** — **FastAPI** backend with structured JSON APIs and a **React + TypeScript** frontend featuring real-time query streaming, source attribution display, and document upload UX.
- **Improved semantic retrieval precision** across structured enterprise documents by building a **FAISS-based vector retrieval infrastructure** with context-aware and heading-aware chunking.
- **Expanded knowledge extraction to audio and video** by architecting **Whisper-powered transcription pipelines** with timestamp-aware chunking, enabling searchable meeting intelligence across recordings and documents.
- **Optimized AI observability and routing efficiency** by integrating **OpenAI APIs**, prompt orchestration, semantic embeddings, and performance analytics across the full inference pipeline.

Distributed Fitness Platform — Spring Boot Microservices + AI Recommendations

- **Architected a scalable microservices fitness platform** using **Spring Boot**, implementing **synchronous REST APIs** and an **asynchronous RabbitMQ** event pipeline for efficient inter-service communication.
- **Enabled independent service scaling and zero-config deployments** by integrating **Netflix Eureka** for service discovery and a centralized **Spring Cloud Config Server** for environment-specific configuration management.
- **Delivered personalized workout recommendations** by building a rule-based **ML microservice** using user activity history, frequency patterns, and fitness goals — served via a dedicated REST endpoint with sub-100ms response time.

Education

Amrita Vishwa Vidyapeetham

Bachelor Of Technology in Computer Science and Engineering

2020 - 2024

7.1/10